

Multimodal Learning for Spatio-Temporal Data Mining

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Abstract

Spatio-temporal data mining (STDM) has become crucial in multimedia, driven by the surge of multimodal data from remote sensing, IoT sensors, social media, surveillance systems, mobile devices, and crowdsourced platforms. Traditional single-modal methods, though successful, struggle to capture real-world complexity. Integrating multiple modalities yields richer, more accurate insights, boosting spatio-temporal analysis. This **half-day tutorial**, *MM4ST: Multimodal Learning for STDM*, offers a comprehensive overview, covering STDM fundamentals, challenges in aligning and fusing heterogeneous data, advanced multimodal modeling techniques, and emerging research directions. Attendees will acquire practical knowledge to develop scalable and robust spatio-temporal mining solutions. All materials will be publicly available online.

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1 Tutorial Outline

MM4ST is structured into four parts, spanning fundamentals, advanced methods, and applications of multimodal STDM.

Part 1: Foundations of STDM: 1) *Background*: Growing complexity of ST problems and AI's role in interpreting, predicting,

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and optimizing dynamic ST patterns; 2) *Development*: Evolution of STDM, sensor technology advancement, and increasing real-world complexity, with key breakthroughs highlighted; 3) *The Era of Large Models*: Transformative role of LLMs and FMs in enabling generalizable, robust ST analytics.

Part 2: Exploring Multimodal ST Data: 1) *Data Modalities*: Key ST data types (geospatial, time-series, sensor) and processing challenges (scalability, noise, missing values); 2) *Data Enrichment*: Integrating complementary sources; how multimodal inputs boost STDM accuracy and context; 3) *Integration Challenges*: Heterogeneous fusion hurdles: spatio-temporal misalignment, quality inconsistency, modality imbalance, cross-modal interaction modeling.

Part 3: Methodology for Multimodal STDM: 1) *Classical Models*: Traditional approaches for multimodal STDM, their limitations, and evolution in handling complex ST relationships; 2) *Multimodal Fusion Techniques*: Advanced fusion paradigms for integrating multimodal ST data, with design/architecture trade-offs; 3) *Real-World Applications*: Case studies in urban region understanding, indicator prediction, ST forecasting, and multimodal time series analysis.

Part 4: Future Directions in Multimodal STDM: Promising directions including multimodal ST foundation models, prompt learning, causal reasoning, interpretable AI, knowledge graphs, and cross-task domain generalization.

2 History & Overlap Statement

Previous Related Tutorials from Our Organizers:

- **WWW'25 – Web-Centric Human Mobility Analytics**: Sydney, Australia (April 28, 2025) – Focus on human mobility analytics (methods, applications, future direction) in the LLM era.
- **KDD'24 – Tutorial on Foundation Models for Time Series** [11]: Barcelona, Spain (August 29, 2024) – 120 onsite attendees.
- **ICDM'23 – Tutorial on Time Series** [27]: Shanghai, China (December 3, 2023) – 30 attendees, focusing on time series analysis from an interdisciplinary perspective.

- **KDD'22 – Tutorial on Time Series** [26]: Washington DC, USA (August 14, 2022) – 70 onsite attendees, 7500+ tutorial website pageviews, focusing on time series and industrial applications.

Previous Related Surveys from Our Organizers:

- Service route and time prediction in instant delivery: TKDE'24 [24].
- Time Series Transformers: IJCAI'23 [28] (Paper Digest Most Influential Paper, 1/700+); Time Series Data Augmentation: IJCAI'21 [25] (Paper Digest Most Influential Paper, 1/600+).
- Self-supervised Learning and LLMs for Time Series: TPAMI'24 [33], IJCAI'24 [8]; Time Series Foundation Models: KDD'24 [11].
- Cross-Domain Data Fusion in Urban Computing: Inf. Fusion'24 [36].
- Spatio-Temporal GNNs for Urban Computing: TKDE'23 [36].

Previous Related Papers from Our Organizers:

- Multimodal Learning for ST Data: AAAI'25 UrbanVLP [6], WWW'25 SatCLE [7], AAAI'25 C2F-TP [23], MM'24 UrbanCross [34], WWW'24 UrbanCLIP [31]; Applications: AAAI'25 Airadar [21], KDD'25 DynST [29], KDD'25 PatchSTG [4], NeurIPS'25 Terra [2], ICDE'25 F-TSL [18], IJCAI'24 DeepPA [32], IJCAI'24 STFNN [5], IJCAI'24 TrajCL [16], AAAI'23 AirFormer [12], SIGMOD'23 LightCTS [10].
- Time Series Analysis: ICLR'24 CARD [22], PVLDB'25 TimeDC [17], ICLR'24 Pathformer [1], ICDE'24 URCL [19], AAAI'23 AirFormer [12], KDD'22 Quatformer [3], ICML'22 FEDformer [35].
- LLMs for Time Series Analysis: AAAI'25 TimeCMA [13], ICLR'24 Time-LLM [9], KDD'24 PeFAD [30], WWW'24 UniTime [14].
- Pre-trained Models for Spatial Time Series: ICLR'24 NuwaDynamics [20], GIS'22 STGCL [15].

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References

- [1] Peng Chen, Yingying Zhang, Yunyao Cheng, Yang Shu, Yihang Wang, Qingsong Wen, Bin Yang, and Chenjuan Guo. 2024. Multi-scale Transformers with Adaptive Pathways for Time Series Forecasting. In *ICLR*.
- [2] Wei Chen, Xixuan Hao, Yuxuan Liang, et al. [n. d.]. Terra: A Multimodal Spatio-Temporal Dataset Spanning the Earth. In *NeurIPS*.
- [3] Weiqi Chen, Wenwei Wang, Bingqing Peng, Qingsong Wen, Tian Zhou, and Liang Sun. 2022. Learning to rotate: Quaternion transformer for complicated periodical time series forecasting. In *KDD*. 146–156.
- [4] Yuchen Fang, Yuxuan Liang, Bo Hui, Zezhi Shao, Liwei Deng, Xu Liu, Xinke Jiang, and Kai Zheng. 2024. Efficient Large-Scale Traffic Forecasting with Transformers: A Spatial Data Management Perspective. *arXiv preprint arXiv:2412.09972* (2024).
- [5] Yutong Feng, Qiongyan Wang, Yutong Xia, Junlin Huang, Siru Zhong, and Yuxuan Liang. 2024. Spatio-temporal field neural networks for air quality inference. In *IJCAI*.
- [6] Xixuan Hao, Wei Chen, Yibo Yan, Siru Zhong, Kun Wang, Qingsong Wen, and Yuxuan Liang. 2025. UrbanVLP: A Multi-Granularity Vision-Language Pre-Trained Foundation Model for Urban Indicator Prediction. (2025).
- [7] Xixuan Hao, Wei Chen, Xingchen Zou, and Yuxuan Liang. 2025. Nature Makes No Leaps: Building Continuous Location Embeddings with Satellite Imagery from the Web. In *WWW*.
- [8] Yushan Jiang, Zijie Pan, Xikun Zhang, Sahil Garg, Anderson Schneider, Yuriy Nevmyvaka, and Dongjin Song. 2024. Empowering Time Series Analysis with Large Language Models: A Survey. In *IJCAI*. 8095–8103.
- [9] Ming Jin, Shiyu Wang, Lintao Ma, Zhixuan Chu, James Y Zhang, Xiaoming Shi, Pin-Yu Chen, Yuxuan Liang, Yuan-Fang Li, et al. 2024. Time-llm: Time series forecasting by reprogramming large language models. In *ICLR*.
- [10] Zhichen Lai, Dalin Zhang, Huan Li, Christian S Jensen, et al. 2023. Lightcts: A lightweight framework for correlated time series forecasting. (2023).
- [11] Yuxuan Liang, Haomin Wen, Yuqi Nie, Yushan Jiang, Ming Jin, et al. 2024. Foundation models for time series analysis: A tutorial and survey. In *KDD*.
- [12] Yuxuan Liang, Yutong Xia, Songyu Ke, Yiwei Wang, Qingsong Wen, Junbo Zhang, Yu Zheng, and Roger Zimmermann. 2023. Airformer: Predicting nationwide air quality in china with transformers. In *AAAI*, Vol. 37. 14329–14337.
- [13] Chenxi Liu, Qianxiong Xu, Hao Miao, Sun Yang, Lingzheng Zhang, Cheng Long, Ziyue Li, and Rui Zhao. 2025. TimeCMA: Towards LLM-Empowered Multivariate Time Series Forecasting via Cross-Modality Alignment. In *AAAI*.
- [14] Xu Liu, Junfeng Hu, Yuan Li, Shizhe Diao, Yuxuan Liang, Bryan Hooi, and Roger Zimmermann. 2024. UniTime: A Language-Empowered Unified Model for Cross-Domain Time Series Forecasting. In *WWW*.
- [15] Xu Liu, Yuxuan Liang, Chao Huang, Yu Zheng, Bryan Hooi, and Roger Zimmermann. 2022. When Do Contrastive Learning Signals Help Spatio-Temporal Graph Forecasting?. In *SIGSPATIAL GIS*.
- [16] Kang Luo, Yuanshao Zhu, Wei Chen, Kun Wang, Zhengyang Zhou, Sijie Ruan, and Yuxuan Liang. 2024. Towards robust trajectory representations: isolating environmental confounders with causal learning. In *IJCAI*.
- [17] Hao Miao, Ziqiao Liu, Yan Zhao, Chenjuan Guo, Bin Yang, Kai Zheng, and Christian S. Jensen. 2025. Less is More: Efficient Time Series Dataset Condensation via Two-fold Modal Matching. *PVLDB* (2025).
- [18] Hao Miao, Ziqiao Liu, Yan Zhao, Kai Zheng, Yupu Zhang, and Christian S. Jensen. 2025. Federated Trajectory Similarity Learning with Privacy-Preserving Clustering. In *ICDE*.
- [19] Hao Miao, Yan Zhao, Chenjuan Guo, Bin Yang, Kai Zheng, Feiteng Huang, Jiandong Xie, and Christian S. Jensen. 2024. A Unified Replay-Based Continuous Learning Framework for Spatio-Temporal Prediction on Streaming Data. In *ICDE*.
- [20] Kun Wang, Hao Wu, Yifan Duan, Guibin Zhang, Kai Wang, Xiaojiang Peng, Yu Zheng, Yuxuan Liang, and Yang Wang. 2024. NuwaDynamics: Discovering and Updating in Causal Spatio-Temporal Modeling. In *ICLR*.
- [21] Qiongyan Wang, Yutong Xia, Siru Zhong, Weichuang Li, Yuankai Wu, Shifen Cheng, et al. 2025. Airadar: Inferring nationwide air quality in china with deep neural networks. In *AAAI*.
- [22] Xue Wang, Tian Zhou, Qingsong Wen, Jinyang Gao, et al. 2024. CARD: Channel Aligned Robust Blend Transformer for Time Series Forecasting. In *ICLR*.
- [23] Zichen Wang, Hao Miao, Senzhang Wang, Renzhi Wang, Jianxin Wang, and Jian Zhang. 2025. C2F-TP: A Coarse-to-Fine Denoising Framework for Uncertainty-aware Trajectory Prediction. In *AAAI*.
- [24] Haomin Wen, Youfang Lin, Lixia Wu, Xiaowei Mao, Tianyue Cai, Yunfeng Hou, Shengnan Guo, et al. 2024. A survey on service route and time prediction in instant delivery: Taxonomy, progress, and prospects. *TKDE* 36, 1 (2024), 1–15.
- [25] Qingsong Wen, Liang Sun, Fan Yang, Xiaomin Song, Jingkun Gao, et al. 2021. Time Series Data Augmentation for Deep Learning: A Survey. In *IJCAI*.
- [26] Qingsong Wen, Linxiao Yang, Tian Zhou, and Liang Sun. 2022. Robust time series analysis and applications: An industrial perspective. In *Tutorial at KDD*.
- [27] Qingsong Wen, Linxiao Yang, Tian Zhou, and Liang Sun. 2023. Robust time series analysis and applications: An industrial perspective. In *Tutorial at ICDM*.
- [28] Qingsong Wen, Tian Zhou, Chaoli Zhang, Weiqi Chen, Ziqing Ma, Junchi Yan, and Liang Sun. 2023. Transformers in time series: A survey. In *IJCAI*. 6778–6786.
- [29] Hao Wu, Haomin Wen, Guibin Zhang, Yutong Xia, Yuxuan Liang, Yu Zheng, et al. 2025. Dynst: Dynamic sparse training for resource-constrained spatio-temporal forecasting. *KDD* (2025).
- [30] Ronghui Xu, Hao Miao, Senzhang Wang, Philip S Yu, and Jianxin Wang. 2024. PeFAD: a parameter-efficient federated framework for time series anomaly detection. In *KDD*. 3621–3632.
- [31] Yibo Yan, Haomin Wen, Siru Zhong, Wei Chen, Haodong Chen, Qingsong Wen, Roger Zimmermann, and Yuxuan Liang. 2024. Urbanclip: Learning text-enhanced urban region profiling with contrastive language-image pretraining from the web. In *the ACM on Web Conference*. 4006–4017.
- [32] Huaiwu Zhang, Yutong Xia, Siru Zhong, Kun Wang, Zekun Tong, Qingsong Wen, et al. 2024. Predicting Carpark Availability in Singapore with Cross-Domain Data: A New Dataset and A Data-Driven Approach. In *IJCAI*.
- [33] Kexin Zhang, Qingsong Wen, Chaoli Zhang, Rongyao Cai, Ming Jin, Yong Liu, James Y Zhang, Yuxuan Liang, et al. 2024. Self-supervised learning for time series analysis: Taxonomy, progress, and prospects. *TPAMI* (2024).
- [34] Siru Zhong, Xixuan Hao, Yibo Yan, Ying Zhang, Yangqiu Song, and Yuxuan Liang. 2024. Urbancross: Enhancing satellite image-text retrieval with cross-domain adaptation. In *ACM MM*. 6307–6315.
- [35] Tian Zhou, Ziqing Ma, Qingsong Wen, Xue Wang, Liang Sun, and Rong Jin. 2022. FEDformer: Frequency enhanced decomposed transformer for long-term series forecasting. In *ICML*.
- [36] Xingchen Zou, Yibo Yan, Xixuan Hao, Yuehong Hu, Haomin Wen, Erdong Liu, Junbo Zhang, Yong Li, et al. 2025. Deep learning for cross-domain data fusion in urban computing: Taxonomy, advances, and outlook. *Information Fusion* (2025).